Evolución del Estado de bienestar de los jóvenes europeos

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Simple Summary: Resumen sencillo para un público no especializado.

Abstract: Resumen técnico del trabajo.

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1. Importación de los DataFrames

```
library(readr)
library(stringr)
library(dplyr)
library(tidyr)
library(magrittr)
library(lubridate)
library(ggplot2)
library(GGally)
```

```
# Cargamos el dataframe
poblacion <- read_tsv('data/poblacion_paises.tsv')

# Limpiamos los nombres de la variable country para obtener los nombres de los paises
colnames(poblacion)[1] <- 'country'
poblacion %<>% mutate(country = sub("^.*,", "", country)) %>% mutate(across(-1, ~
 ↪  as.numeric(gsub('[^0-9.]', '', .x))))

# Calculamos la media poblacional de cada pais desde el 2000 hasta hoy
poblacion %<>% transmute(country, population = rowMeans(select(., -country), na.rm =
 ↪  TRUE))

# Ordenamos los paises de mayor a menor poblacion
# Eliminamos algunos que no pertenecen politicamente a Europa
poblacion %<>% arrange(desc(population)) %>% filter(!(country %in% c('RU', 'UA',
 ↪  'DE_TOT', 'TR', 'FX', 'BY', 'AZ', 'UK')))

# Nos quedamos con los nombres de los 10 primeros
mas.poblacion10 <- poblacion$country[1:10]

# Nos quedamos con los nombres de los 5 primeros
mas.poblacion5 <- poblacion$country[1:5]
```

```
# Cargamos el dataframe
indep.resid <- read_tsv('data/edad_independencia_residencial.tsv')

# La primera columna viene con los valores de sex y country, por lo que hay que
 ↪  separarlos
colnames(indep.resid)[1] <- 'sex.country'
indep.resid %<>% mutate(sex.country = sub("^.*,([^,]*,[^,]*)$", "\\1", sex.country))
 ↪  %>% separate(sex.country, into = c('sex', 'country'), sep = ',') %>% mutate(sex =
 ↪  factor(sex))
```

```r
# Aqui pasamos los valores de edad de emancipacion en funcion de año a formato tidy
# Filtramos para quedarnos con sex = 'T', el total, ya que no nos interesa la
↪  diferenciacion por sexos
# Nos quedamos con los 10 paises mas poblados
indep.resid %<>% mutate(across(c(-1, -2), ~ as.numeric(gsub('[^0-9.]', '', .x)))) %>%
↪  pivot_longer(cols = c(-1, -2), names_to = 'year', values_to = 'age.indep') %>%
↪  mutate(year = as.numeric(year)) %>% filter(sex == 'T' & country %in%
↪  mas.poblacion10) %>% select(-sex) %>% mutate(country = factor(country, levels =
↪  mas.poblacion10, labels = mas.poblacion10))
```

```r
precio.vivienda <- read_tsv('data/precio_vivienda.tsv')

colnames(precio.vivienda)[1] <- 'country'
precio.vivienda %<>% mutate(country = sub(".*,", "", country))

precio.vivienda %<>% mutate(across(-1, ~ as.numeric(gsub('[^0-9.]', '', .x)))) %>%
↪  pivot_longer(cols = -1, names_to = 'year.quadri', values_to =
↪  'rel.price2015.house')

# Como el indice viene por cuatrimestres, nos quedamos con 1 solo valor, el
↪  correspondiente al cuatrimestre 1
precio.vivienda %<>% separate(year.quadri, into = c('year', 'quadri'), sep = '-') %>%
↪  mutate(quadri = factor(quadri, ordered=TRUE)) %>% mutate(year = as.numeric(year))
↪  %>% filter(country %in% mas.poblacion10 & quadri == 'Q1') %>% select(-quadri) %>%
↪  mutate(country = factor(country, levels = mas.poblacion10, labels =
↪  mas.poblacion10))
```

```r
precio.alquiler <- read_tsv('data/precio_alquiler.tsv')

colnames(precio.alquiler)[1] <- 'country'
precio.alquiler %<>% mutate(country = sub(".*,", "", country))

precio.alquiler %<>% mutate(across(-1, ~ as.numeric(gsub('[^0-9.]', '', .x)))) %>%
↪  pivot_longer(cols = -1, names_to = 'year', values_to = 'rel.price2015.rent') %>%
↪  mutate(year = as.numeric(year)) %>% filter(country %in% mas.poblacion10) %>%
↪  mutate(country = factor(country, levels = mas.poblacion10, labels =
↪  mas.poblacion10))
```

```r
salarios2 <- read_tsv('data/salarios2.tsv')

# Este dataframe ha sido obtenido de otra fuente
# Por eso los nombres de los paises vienen en otro formato y hay que hacer el filtrado
↪  manualmente otra vez
salarios2 %<>% transmute(country = ref_area.label, year = time, earnings =
↪  0.9*obs_value, source = factor(source.label))
names.country <- c('Germany', 'France', 'Italy', 'Spain', 'Poland', 'Romania',
↪  'Netherlands', 'Belgium', 'Greece', 'Czechia')
names.country %in% salarios2$country
salarios2 %<>% filter(country %in% names.country) %>% mutate(country = factor(country,
↪  levels = names.country, labels = mas.poblacion10))

# En algunos paises se ha cambiado el metodo de cuantificar los salarios, por lo que
↪  hay saltos muy grandes de un año a otro
```

```r
# Identificamos los paises en los que ha pasado esto
major.source <- salarios2 %>% count(country, source) %>% group_by(country) %>%
 ↪  arrange(desc(n)) %>% slice(1) %>% mutate(major.source = source) %>%
 ↪  select(c(-source, -n))

# Para no estropear la tendencia temporal, nos quedamos con los datos correspondientes
 ↪  a 1 solo metodo, del que mas observaciones haya
salarios2 %<>% full_join(major.source) %>% mutate(earnings = ifelse(source ==
 ↪  major.source, earnings, NA)) %>% select(c(-major.source, -source))
```

2. Datos ausentes

```r
# Juntamos todas las variables en un dataframe por paises y años
jovenes.df <- indep.resid %>% full_join(salarios2) %>% full_join(precio.alquiler) %>%
 ↪  full_join(precio.vivienda)
```

```
## Joining with `by = join_by(country, year)`
## Joining with `by = join_by(country, year)`
## Joining with `by = join_by(country, year)`
```

```r
dim(jovenes.df)
```

```
## [1] 269   6
```

```r
# Numero de registros y de variables

vars.num <- setdiff(colnames(jovenes.df)[sapply(jovenes.df, is.numeric)], 'year')
vars.num
```

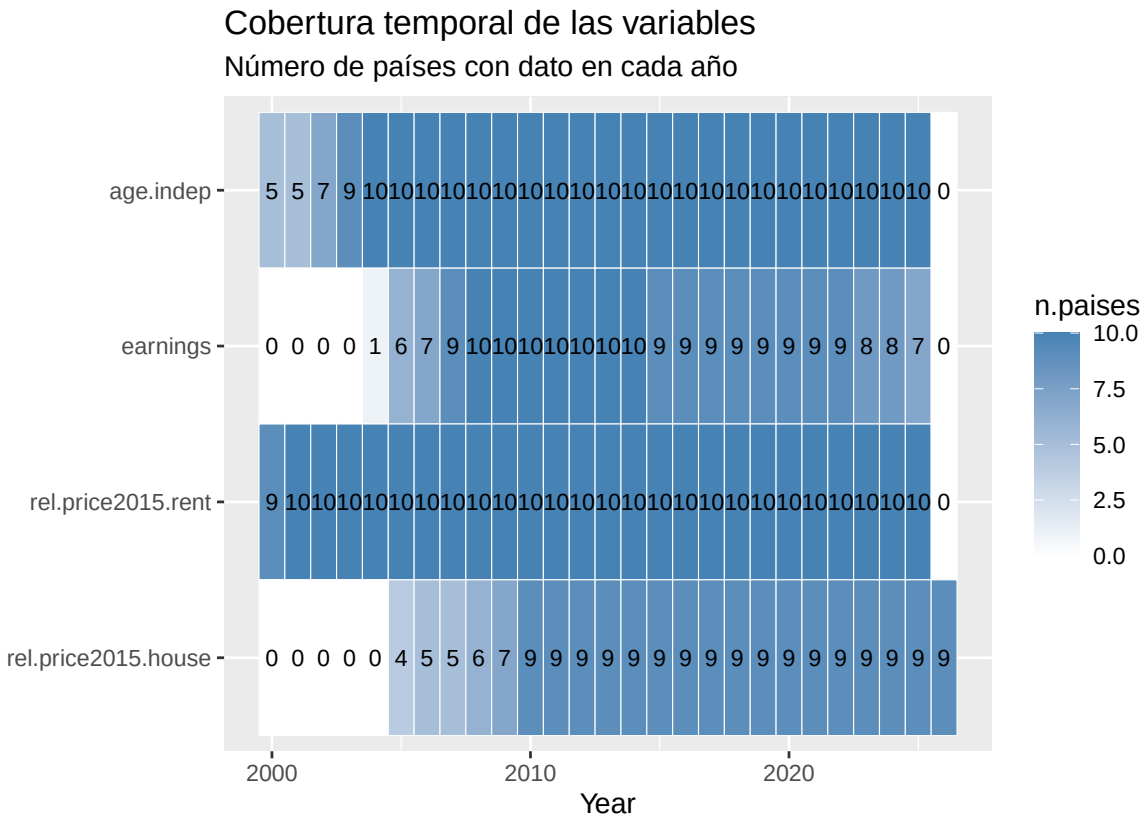
```
## [1] "age.indep"           "earnings"            "rel.price2015.rent"
## [4] "rel.price2015.house"
```

```r
# Variables cuantitativas continuas. 'country' es cualitativa nominal y 'year' es
 ↪  cuantitativa discreta (indice temporal)

por.var <- function(f) unname(sapply(jovenes.df[vars.num], f))
# Aplica una funcion a cada variable numerica y devuelve un vector sin nombres
```

```r
cobertura <- jovenes.df %>% group_by(year) %>% summarise(across(all_of(vars.num), ~
 ↪  sum(!is.na(.x)))) %>% pivot_longer(-year, names_to = 'variable', values_to =
 ↪  'n.paises') %>% mutate(variable = factor(variable, levels = rev(vars.num)))

ggplot(cobertura, aes(x = year, y = variable, fill = n.paises)) + geom_tile(color =
 ↪  'white') + geom_text(aes(label = n.paises), size = 3) + scale_fill_gradient(low =
 ↪  'white', high = 'steelblue', limits = c(0, length(unique(jovenes.df$country)))) +
 ↪  labs(title = 'Cobertura temporal de las variables', subtitle = 'Número de países
 ↪  con dato en cada año') + xlab('Year') + ylab('')
```




```
ggsave('results/univ_cobertura.pdf', width = 14, height = 10, dpi = 300)
```

```
# Limitamos el estudio desde el 2010 hasta el 2025
jovenes.df %<>% filter(year >= 2010 & year <= 2025)

# Obtenemos el primer año en el que comienzan las medidas de salarios y precios
# Tomamos este año como valor de referencia para poder compararlos luego mas
 ↪  facilmente
year.start <- jovenes.df %>% group_by(country) %>% arrange(year) %>% select(c(country,
 ↪  year, rel.price2015.rent, rel.price2015.house, earnings)) %>%
 ↪  filter(if_all(everything(), ~ !is.na(.x))) %>% slice(1) %>% ungroup() %>%
 ↪  transmute(country, year.start = year)

jovenes.df <- full_join(jovenes.df, year.start)
```

```
## Joining with `by = join_by(country)`
```

```
# Ahora tomamos, para cada variable, un valor de 100 en el año de referencia, asi las
 ↪  podremos comparar mejor graficamente
jovenes.df %<>% group_by(country) %>% mutate(earnings = earnings/earnings[year ==
 ↪  unique(year.start)]*100, price.rent = rel.price2015.rent/rel.price2015.rent[year
 ↪  == unique(year.start)]*100, price.house =
 ↪  rel.price2015.house/rel.price2015.house[year == unique(year.start)]*100) %>%
 ↪  select(c(-year.start, -rel.price2015.rent, -rel.price2015.house)) %>% ungroup()

# Calculamos los salarios en proporcion al precio de la vivienda y a los alquileres,
 ↪  respectivamente
jovenes.df %<>% mutate(earn.prop.house = earnings/price.house*100, earn.prop.rent =
 ↪  earnings/price.rent*100)
```

```r
# Calculamos el aumento porcentual de los salarios, alquileres y precio de la vivienda
↪  para cada año
jovenes.df %<>%
  group_by(country) %>%
  arrange(year, .by_group = TRUE) %>%
  mutate(
    age.diff = age.indep - lag(age.indep),
    earn.diff = earnings - lag(earnings),
    house.diff = price.house - lag(price.house),
    rent.diff = price.rent - lag(price.rent)
  ) %>%
  ungroup()
```

```r
# Calculamos ahora cuales son las variables cuantitativas continuas tras las
↪  transformaciones
vars.num <- setdiff(colnames(jovenes.df)[sapply(jovenes.df, is.numeric)], 'year')
vars.num
```

```
##  [1] "age.indep"       "earnings"        "price.rent"      "price.house"
##  [5] "earn.prop.house" "earn.prop.rent"  "age.diff"        "earn.diff"
##  [9] "house.diff"      "rent.diff"
```

3. Estudio de variables

```r
sumario.jovenes <- data.frame(variable = colnames(jovenes.df))
sumario.jovenes$type <- sapply(jovenes.df, function(x) class(x)[1])
sumario.jovenes$levels <- sapply(jovenes.df, function(x) length(unique(x)))
sumario.jovenes$topLevel <- sapply(jovenes.df, function(x) names(which.max(table(x))))
sumario.jovenes$topCount <- sapply(jovenes.df, function(x) max(table(x)))
sumario.jovenes$topFrac <- sumario.jovenes$topCount / nrow(jovenes.df)
sumario.jovenes$missFreq <- sapply(jovenes.df, function(x) sum(is.na(x)))
sumario.jovenes$missFrac <- sumario.jovenes$missFreq / nrow(jovenes.df)

sumario.jovenes
```

```
##           variable    type levels          topLevel topCount topFrac missFreq
## 1          country  factor     10                DE       16 0.10000        0
## 2             year numeric     16              2010       10 0.06250        0
## 3        age.indep numeric     61              23.7       12 0.07500        0
## 4         earnings numeric    128               100        9 0.05625       25
## 5       price.rent numeric    136               100        9 0.05625       16
## 6      price.house numeric    136               100        9 0.05625       16
## 7  earn.prop.house numeric    128               100        9 0.05625       25
## 8   earn.prop.rent numeric    128               100        9 0.05625       25
## 9         age.diff numeric     29                 0       31 0.19375       10
## 10       earn.diff numeric    127 -26.0178668438882        1 0.00625       34
## 11      house.diff numeric    134  1.45547945205479        3 0.01875       25
## 12       rent.diff numeric    132 0.210084033613441        2 0.01250       25
##    missFrac
## 1   0.00000
## 2   0.00000
## 3   0.00000
## 4   0.15625
## 5   0.10000
```

```
## 6   0.10000
## 7   0.15625
## 8   0.15625
## 9   0.06250
## 10  0.21250
## 11  0.15625
## 12  0.15625
```

```
cobertura.var <- data.frame(variable = vars.num)
cobertura.var$year.min <- unname(sapply(vars.num, function(v)
 ↪  min(jovenes.df$year[!is.na(jovenes.df[[v]])])))
cobertura.var$year.max <- unname(sapply(vars.num, function(v)
 ↪  max(jovenes.df$year[!is.na(jovenes.df[[v]])])))
cobertura.var$n.obs <- unname(sapply(vars.num, function(v)
 ↪  sum(!is.na(jovenes.df[[v]]))))
cobertura.var$n.paises <- unname(sapply(vars.num, function(v)
 ↪  length(unique(jovenes.df$country[!is.na(jovenes.df[[v]])]))))
cobertura.var
```

```
##           variable year.min year.max n.obs n.paises
## 1        age.indep     2010     2025   160       10
## 2         earnings     2010     2025   135        9
## 3       price.rent     2010     2025   144        9
## 4      price.house     2010     2025   144        9
## 5  earn.prop.house     2010     2025   135        9
## 6   earn.prop.rent     2010     2025   135        9
## 7         age.diff     2011     2025   150       10
## 8        earn.diff     2011     2025   126        9
## 9       house.diff     2011     2025   135        9
## 10       rent.diff     2011     2025   135        9
```

4. Análisis univariante

4.1. Estadísticos descriptivos de las variables numéricas

```
mystats <- function(x, na.omit = TRUE) {
  if (na.omit) x <- x[!is.na(x)]
  m <- mean(x)
  n <- length(x)
  s <- sd(x)
  skew <- sum((x - m)^3 / s^3) / n
  kurt <- sum((x - m)^4 / s^4) / n - 3
  return(c(n = n, mean = m, trimmed = mean(x, trim = 0.1), median = median(x), min =
   ↪  min(x), q1 = quantile(x, 0.25, names = FALSE), q3 = quantile(x, 0.75, names =
   ↪  FALSE), max = max(x), stdev = s, var = var(x), mad = mad(x), IQR = IQR(x), skew
   ↪  = skew, kurtosis = kurt))
}
# Funcion del Tema 6 (asimetria y curtosis en exceso), ampliada con media recortada
 ↪  (10%), cuartiles, rango intercuartilico y MAD

estad <- sapply(jovenes.df[vars.num], mystats)

t(round(estad[c('n', 'mean', 'trimmed', 'median', 'min', 'q1', 'q3', 'max'), ], 2))
```

```
##                  n   mean trimmed median    min     q1     q3    max
## age.indep      160  26.77   26.74  27.00  23.00  23.90  29.00  30.90
```

```
## earnings        135 121.47  109.75 103.22  64.26  95.58 118.49 431.60
## price.rent      144 115.02  111.73 108.71  99.46 104.36 118.98 190.88
## price.house     144 116.87  111.63 106.05  71.00  94.69 130.36 247.48
## earn.prop.house 135 106.72   98.67 100.00  63.85  83.69 112.76 330.68
## earn.prop.rent  135 103.76   96.97  97.48  64.35  86.01 105.86 258.22
## age.diff        150   0.02    0.02   0.00  -1.80  -0.10   0.20   1.80
## earn.diff       126   6.04    3.72   4.08 -26.02  -3.03  10.20  64.47
## house.diff      135   4.20    3.71   3.47 -13.68   0.46   6.87  43.17
## rent.diff       135   2.86    2.23   1.81  -1.30   1.15   3.17  19.45
```

```
# Tendencia central y posicion

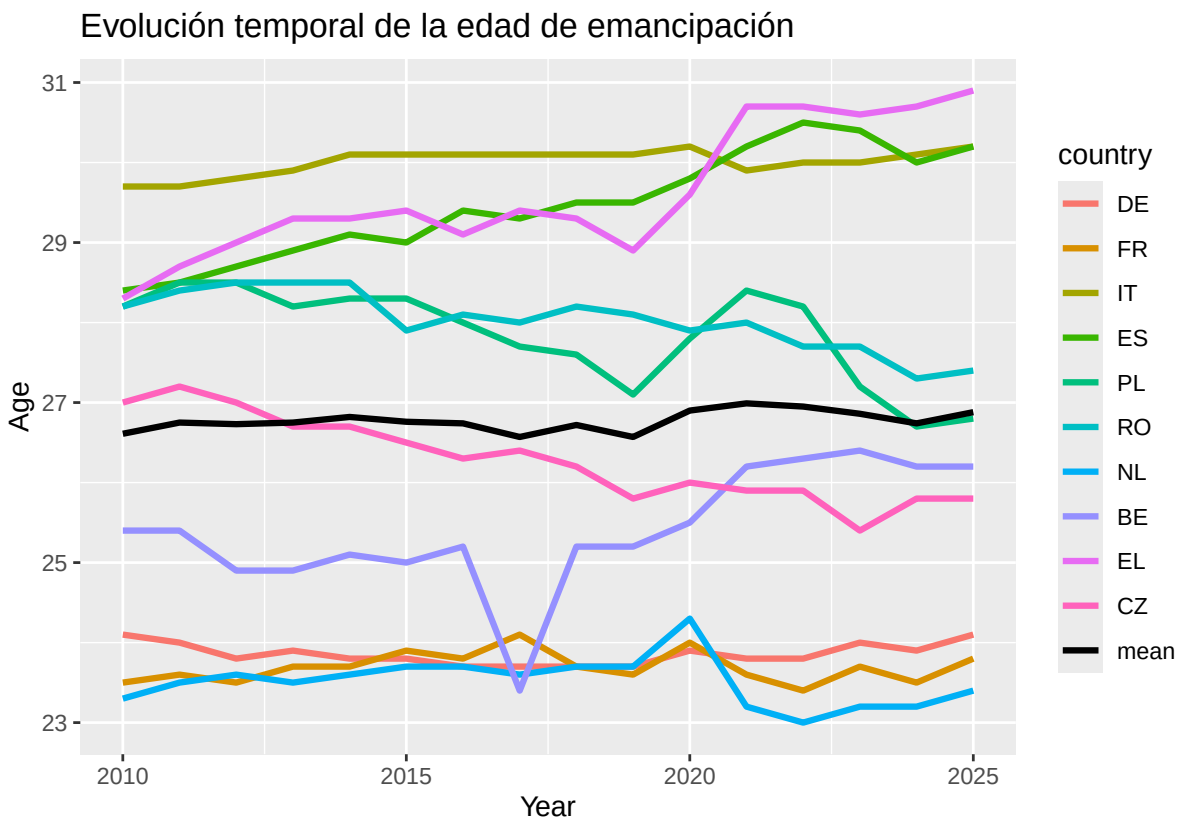
t(round(estad[c('stdev', 'var', 'mad', 'IQR', 'skew', 'kurtosis'), ], 2))
```

```
##                 stdev     var   mad   IQR  skew kurtosis
## age.indep        2.49    6.21  3.56  5.10 -0.03    -1.48
## earnings        55.57 3087.57 15.46 22.91  3.03    10.57
## price.rent      17.18  295.04  8.92 14.62  2.06     4.55
## price.house     34.21 1170.42 22.36 35.67  1.48     2.01
## earn.prop.house 42.96 1845.74 22.56 29.07  3.11    10.71
## earn.prop.rent  33.83 1144.40 15.12 19.85  2.74     8.00
## age.diff         0.35    0.12  0.30  0.30 -0.18     8.67
## earn.diff       16.95  287.13  9.41 13.24  1.68     3.61
## house.diff       7.66   58.70  4.87  6.41  1.35     5.13
## rent.diff        3.31   10.96  1.31  2.02  2.63     8.27
```

```
# Dispersion y forma (skew > 0: cola larga a la derecha; kurtosis > 0: colas mas
↪  pesadas que la normal)
```

4.2. Edad de emancipación

```
age.mean <- jovenes.df %>% group_by(year) %>% summarise(age.mean = mean(age.indep,
↪  na.rm = TRUE), .groups = "drop")
ggplot() + geom_line(data = jovenes.df, mapping = aes(x = year, y = age.indep, color
↪  = country), linewidth = 1.15, na.rm = TRUE) + geom_line(data = age.mean, mapping =
↪  aes(x = year, y = age.mean, color = 'mean'), linewidth = 1.15, na.rm = TRUE) +
  scale_color_manual(values =
   ↪  c(setNames(scales::hue_pal()(length(unique(jovenes.df$country))),
   ↪  unique(jovenes.df$country)), "mean" = "black")) +
  labs(title = 'Evolución temporal de la edad de emancipación') + xlab('Year') +
   ↪  ylab('Age')
```




```
ggsave('results/edad_emancipacion.pdf', width = 14, height = 10, dpi = 300)
```

```
media.edad <- jovenes.df %>% group_by(country) %>% summarise(n.years = n(), media =
 ↪  round(mean(age.indep, na.rm = TRUE),2), media.pct = round(mean(age.diff, na.rm =
 ↪  TRUE),2), total.dif = age.indep[length(age.indep)]-age.indep[1], .groups = 'drop')
 ↪  %>% as.data.frame()
media.edad
```

```
##    country n.years media media.pct total.dif
## 1       DE      16 23.86      0.00       0.0
## 2       FR      16 23.69      0.02       0.3
## 3       IT      16 30.01      0.03       0.5
## 4       ES      16 29.46      0.12       1.8
## 5       PL      16 27.84     -0.09      -1.4
## 6       RO      16 28.02     -0.05      -0.8
## 7       NL      16 23.51      0.01       0.1
## 8       BE      16 25.41      0.05       0.8
## 9       EL      16 29.62      0.17       2.6
## 10      CZ      16 26.29     -0.08      -1.2
```

```
# Media anual de cada pais de la edad de emancipación
```

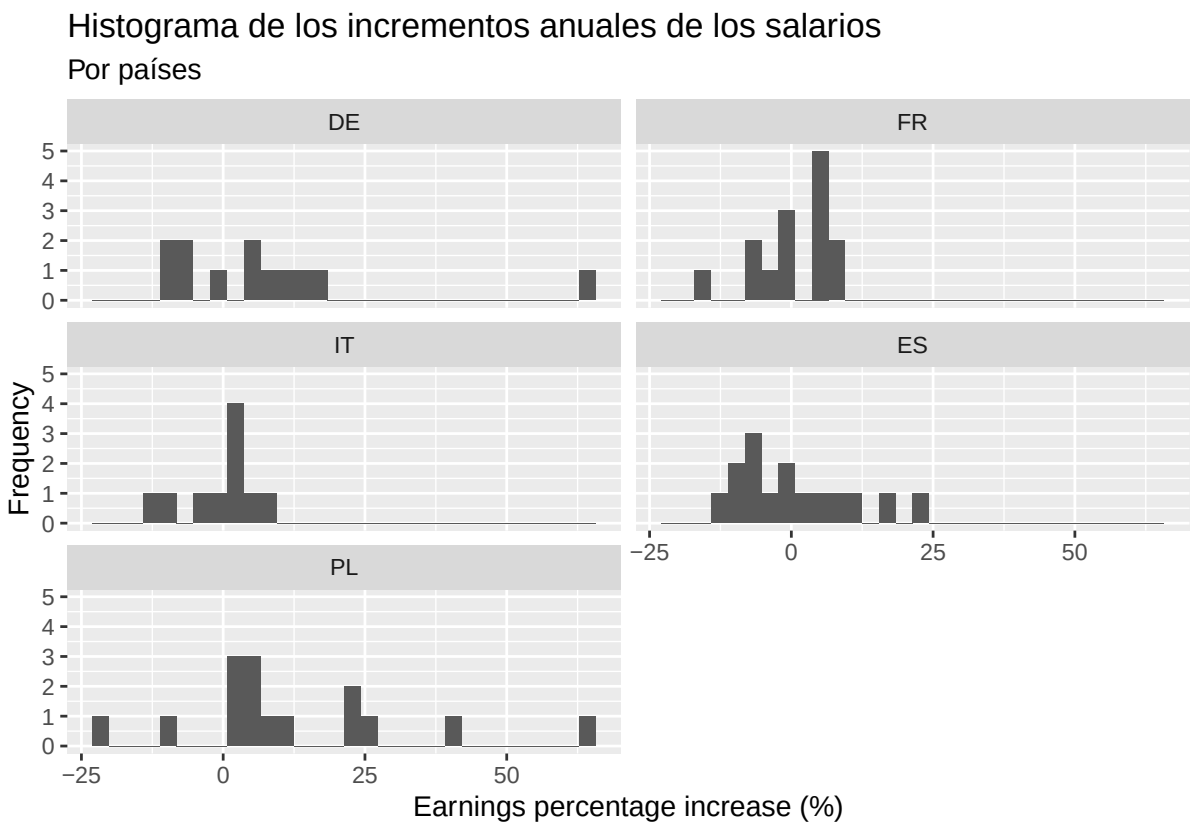
4.3. Media del aumento anual por país: salarios, alquiler y vivienda

```
## year.ini year.fin
##     2010     2020
```

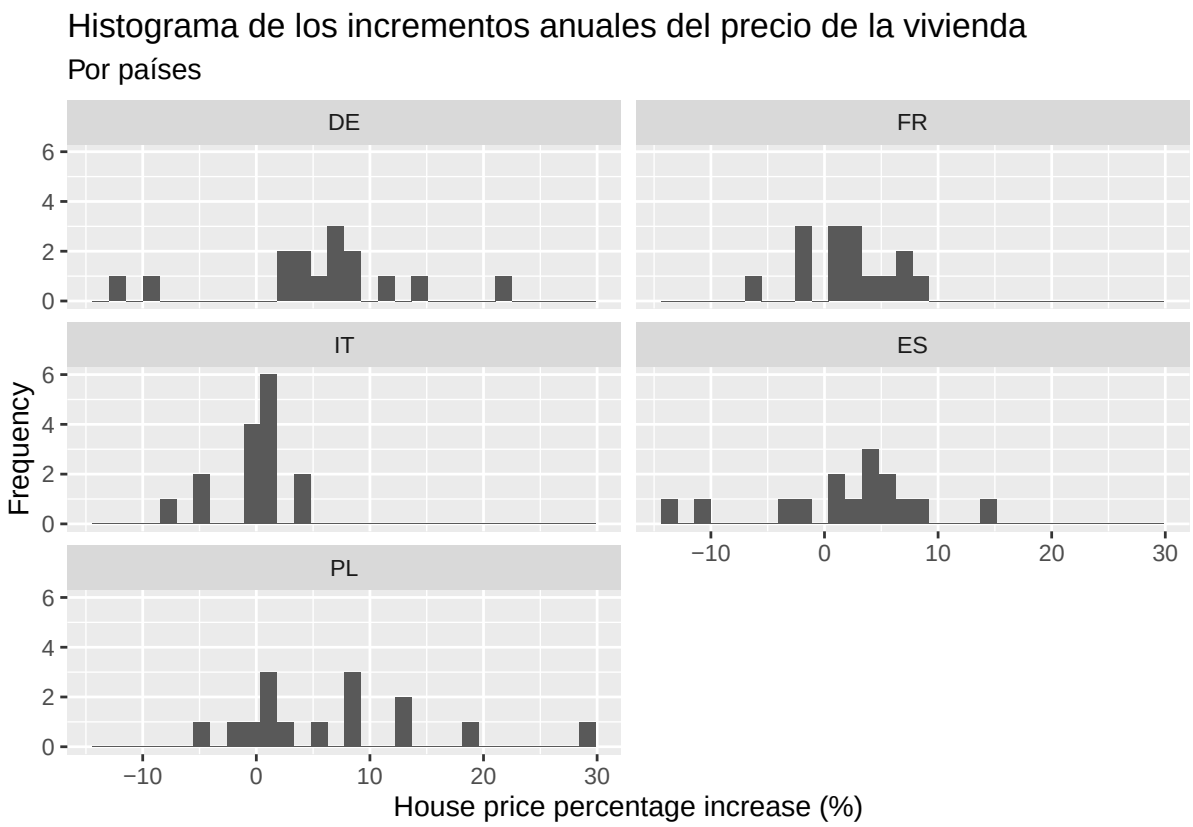
```
##   country  n
## 1      DE 10
```

```
## 2      FR 10
## 3      IT 10
## 4      ES 10
## 5      PL 10
## 6      RO 10
## 7      NL 10
## 8      BE 10
## 9      CZ 10
```

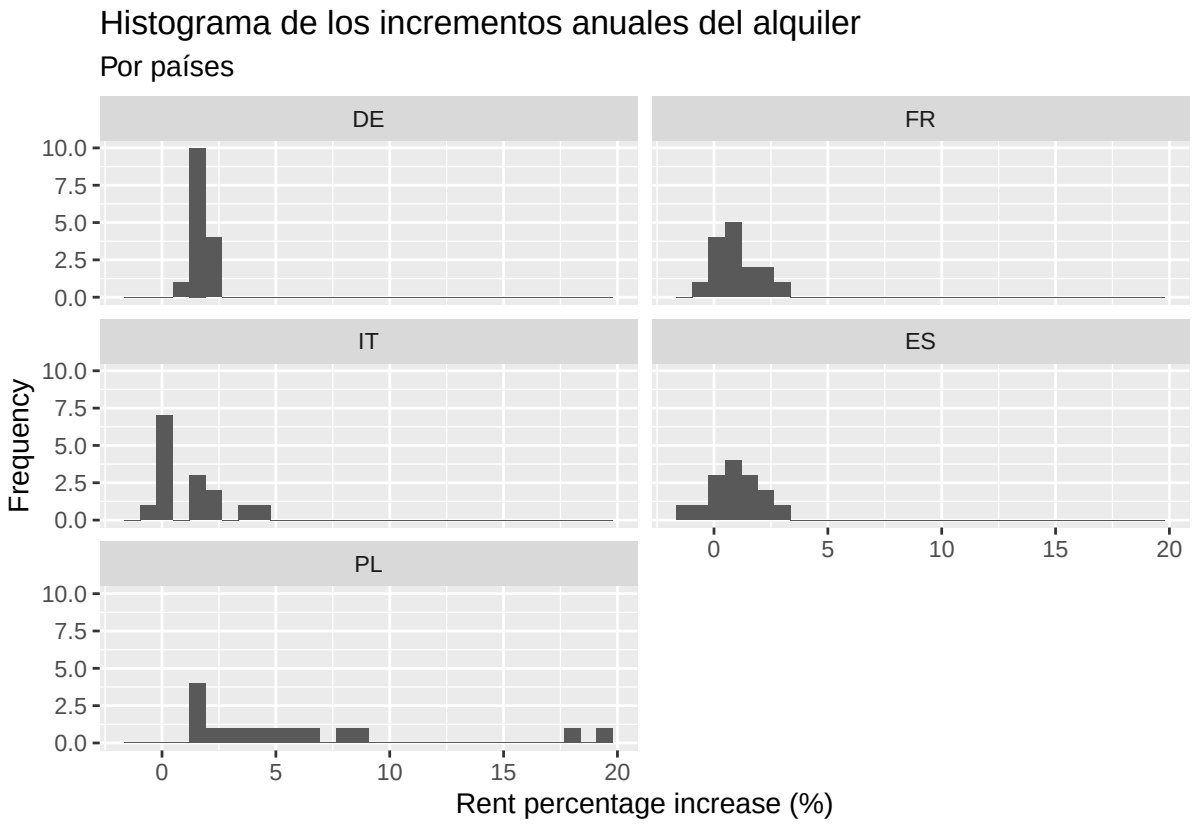
```
jovenes.df %>% filter(country %in% mas.poblacion5) %>% ggplot(aes(x=earn.diff)) +
↪  geom_histogram(bins = 30, na.rm = TRUE) + facet_wrap(~country, nrow = 3)+
↪  xlab('Earnings percentage increase (%)') + ylab('Frequency') + labs(title =
↪  'Histograma de los incrementos anuales de los salarios', subtitle = 'Por países')
```




```
jovenes.df %>% filter(country %in% mas.poblacion5) %>% ggplot(aes(x=house.diff)) +
↪  geom_histogram(bins = 30, na.rm = TRUE) + facet_wrap(~country, nrow = 3)+
↪  xlab('House price percentage increase (%)') + ylab('Frequency') + labs(title =
↪  'Histograma de los incrementos anuales del precio de la vivienda', subtitle = 'Por
↪  países')
```




```
jovenes.df %>% filter(country %in% mas.poblacion5) %>% ggplot(aes(x=rent.diff)) +
↪  geom_histogram(bins = 30, na.rm = TRUE) + facet_wrap(~country, nrow = 3)+
↪  xlab('Rent percentage increase (%)') + ylab('Frequency') + labs(title =
↪  'Histograma de los incrementos anuales del alquiler', subtitle = 'Por países')
```



```r
medias.pais <- aum.comun %>% group_by(country) %>% summarise(n.years = n(),
 ↪  across(all_of(vars.pct), mean), .groups = 'drop') %>% mutate(earn.menos.house =
 ↪  earn.pct - house.pct, earn.menos.rent = earn.pct - rent.pct) %>% as.data.frame()
medias.pais %>% mutate(across(-c(country, n.years), ~ round(.x, 2)))
```

```
##   country n.years house.pct rent.pct earn.pct earn.menos.house earn.menos.rent
## 1      DE      10      6.12     1.47     8.28             2.16            6.81
## 2      FR      10      1.78     0.73    -0.09            -1.87           -0.83
## 3      IT      10     -1.49     0.65    -0.34             1.15           -0.99
## 4      ES      10     -0.53     0.39    -1.13            -0.60           -1.52
## 5      PL      10      2.59     3.15     6.69             4.10            3.53
## 6      RO      10      0.46     1.76    13.49            13.04           11.73
## 7      NL      10      2.47     2.99     0.13            -2.35           -2.87
## 8      BE      10      2.79     1.44     0.31            -2.48           -1.13
## 9      CZ      10      5.65     2.66     4.36            -1.29            1.70
```

```r
# Media anual de cada pais en la ventana comun y diferencias entre variables
```

```r
pares <- list(c('earn.pct', 'house.pct'), c('earn.pct', 'rent.pct'))

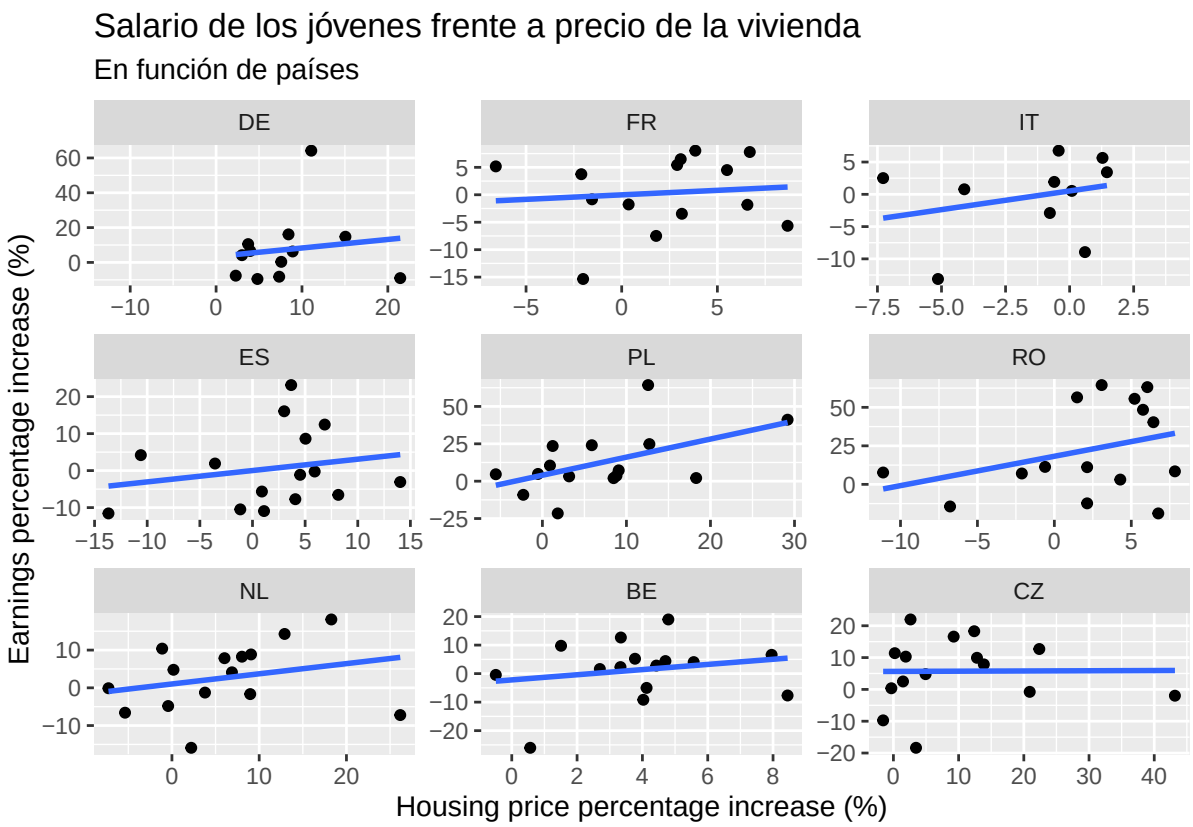
data.frame(par = sapply(pares, paste, collapse = ' vs '), paises.mayor = sapply(pares,
 ↪  function(p) sum(medias.pais[[p[1]]] > medias.pais[[p[2]]])), n.paises =
 ↪  nrow(medias.pais))
```

```
##                     par paises.mayor n.paises
## 1 earn.pct vs house.pct            4        9
## 2  earn.pct vs rent.pct            4        9
```

```r
# paises.mayor: numero de paises en los que la media de la primera variable es mayor
 ↪  que la de la segunda
```

5. Análisis bivariante

```r
jovenes.df %>% filter(country != 'EL') %>% ggplot(aes(x = house.diff, y = earn.diff))
 ↪  + geom_point(na.rm = TRUE) +
   geom_smooth(method = "lm", se = FALSE, na.rm = TRUE) + facet_wrap(.~country, nrow =
    ↪  4, scales = 'free') + ylab('Earnings percentage increase (%)') + xlab('Housing
    ↪  price percentage increase (%)') + labs(title = 'Salario de los jóvenes frente a
    ↪  precio de la vivienda', subtitle = 'En función de países')
```




```
jovenes.df %>% filter(country != 'EL') %>% group_by(country) %>% summarise(cor.pearson
↪  = cor(earn.diff, house.diff, method = 'pearson', use = 'complete.obs'),
↪  cor.spearman = cor(earn.diff, house.diff, method = 'spearman', use =
↪  'complete.obs'))
```

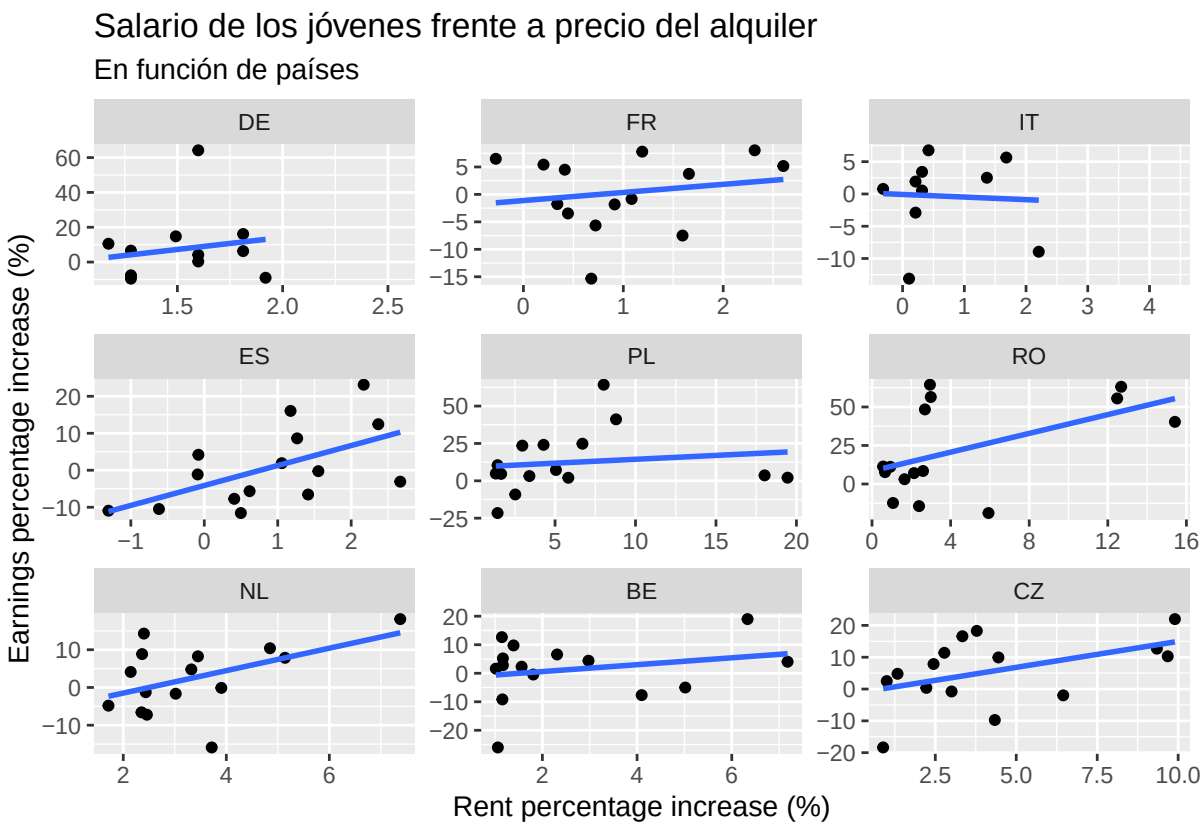
```
## # A tibble: 9 x 3
##   country cor.pearson cor.spearman
##   <fct>         <dbl>        <dbl>
## 1 DE            0.136        0.217
## 2 FR            0.101        0.134
## 3 IT            0.272        0.345
## 4 ES            0.206        0.232
## 5 PL            0.517        0.386
## 6 RO            0.344        0.175
## 7 NL            0.260        0.3
## 8 BE            0.205        0.207
## 9 CZ            0.00748      0.104
```

```
jovenes.df %>% filter(country != 'EL') %>% group_by(country) %>% summarise(slope =
↪  lm(earn.diff ~ house.diff)[[1]][2])
```

```
## # A tibble: 9 x 2
##   country   slope
##   <fct>     <dbl>
## 1 DE      0.489
## 2 FR      0.164
## 3 IT      0.579
## 4 ES      0.307
## 5 PL      1.21
```

```
## 6 RO       1.91
## 7 NL       0.270
## 8 BE       0.905
## 9 CZ       0.00674
```

```
jovenes.df %>% filter(country != 'EL') %>% ggplot(aes(x = rent.diff, y = earn.diff)) +
↪  geom_point(na.rm = TRUE) +
 geom_smooth(method = "lm", se = FALSE, na.rm = TRUE) + facet_wrap(.~country, nrow =
  ↪  4, scales = 'free') + ylab('Earnings percentage increase (%)') + xlab('Rent
  ↪  percentage increase (%)') + labs(title = 'Salario de los jóvenes frente a precio
  ↪  del alquiler', subtitle = 'En función de países')
```



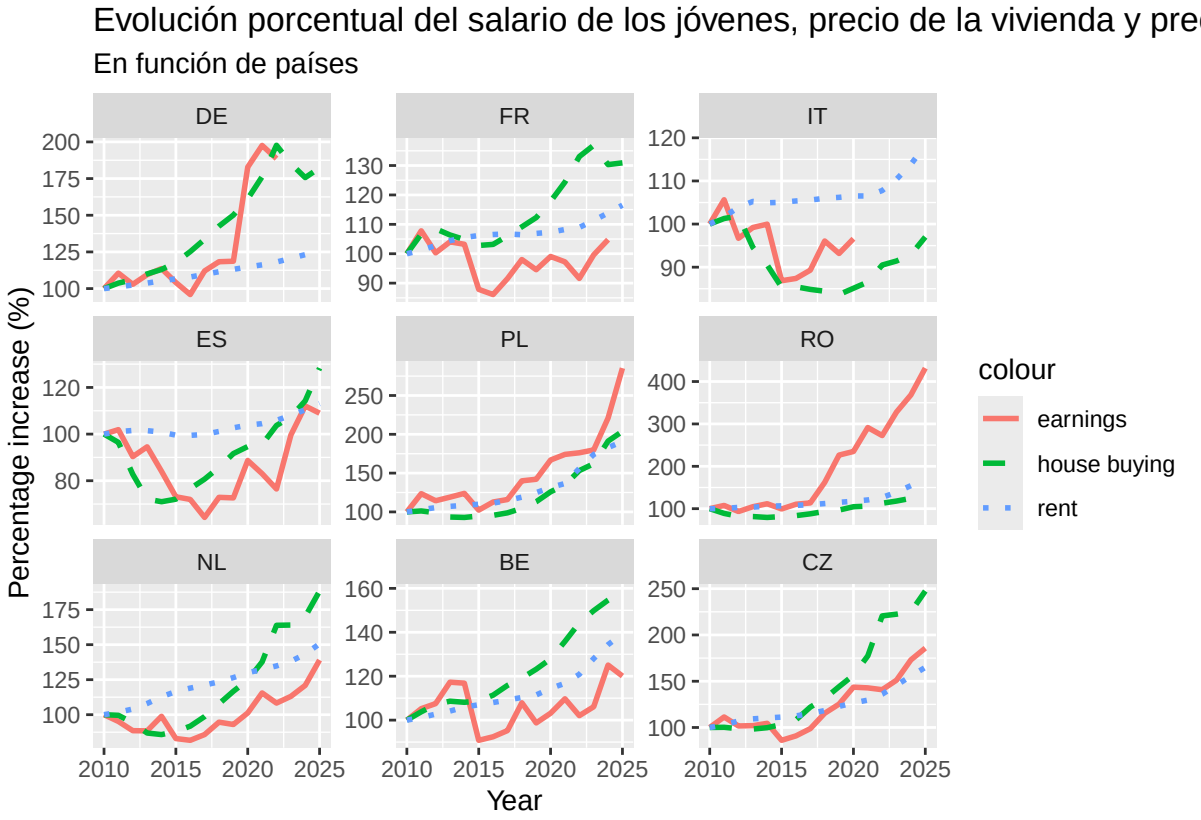

```
jovenes.df %>% filter(country != 'EL') %>% group_by(country) %>% summarise(cor.pearson
↪  = cor(earn.diff, rent.diff, method = 'pearson', use = 'complete.obs'),
↪  cor.spearman = cor(earn.diff, rent.diff, method = 'spearman', use =
↪  'complete.obs'))
```

```
## # A tibble: 9 x 3
##   country cor.pearson cor.spearman
##   <fct>         <dbl>        <dbl>
## 1 DE           0.174        0.154
## 2 FR           0.178        0.0901
## 3 IT          -0.0514       0.354
## 4 ES           0.572        0.554
## 5 PL           0.142        0.236
## 6 RO           0.506        0.439
## 7 NL           0.484        0.346
## 8 BE           0.237        0.211
## 9 CZ           0.470        0.479
```

```r
jovenes.df %>% filter(country != 'EL') %>% group_by(country) %>% summarise(slope =
 ↪  lm(earn.diff ~ rent.diff)[[1]][2])
```

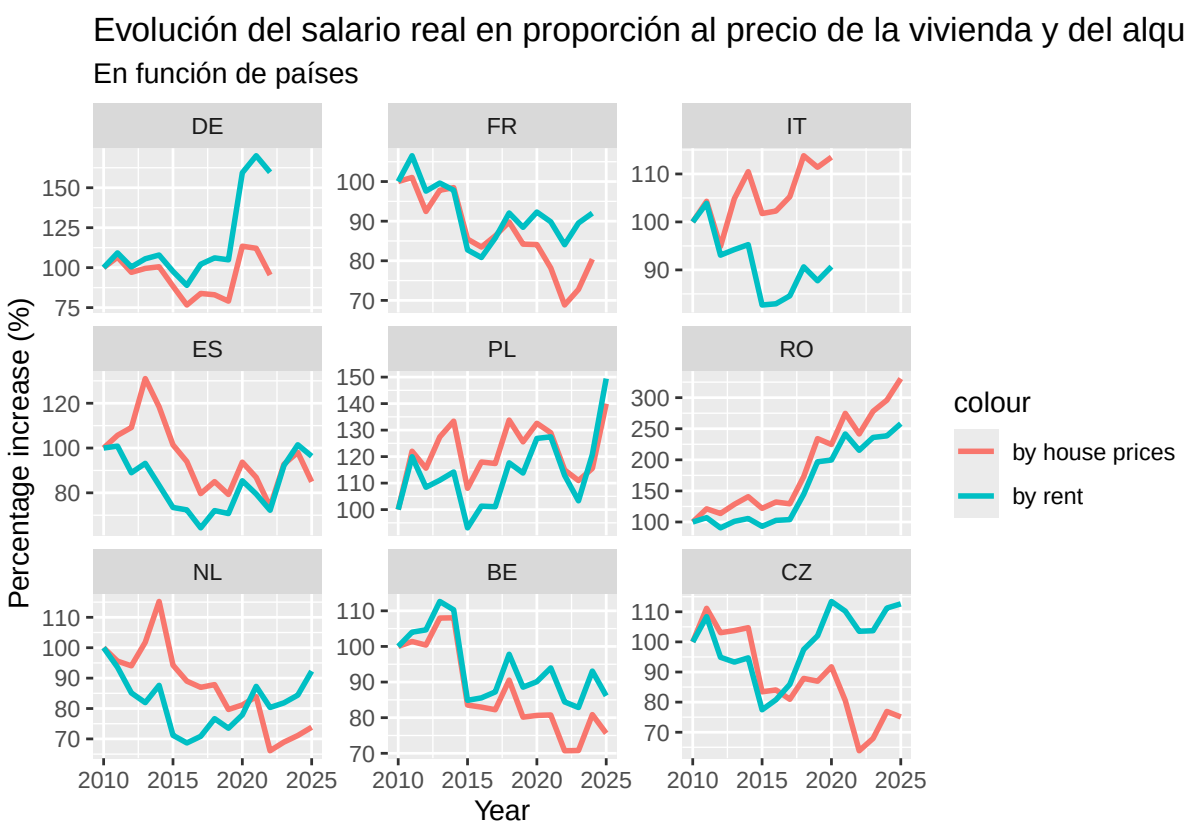
```
## # A tibble: 9 x 2
##   country  slope
##   <fct>    <dbl>
## 1 DE      13.8
## 2 FR       1.48
## 3 IT      -0.401
## 4 ES       5.40
## 5 PL       0.519
## 6 RO       3.05
## 7 NL       2.97
## 8 BE       1.21
## 9 CZ       1.63
```

```r
ggplot(data = filter(jovenes.df, country != 'EL', year >= 2005 & year <= 2025)) +
  geom_line(aes(x = year, y = earnings, color = 'earnings'), linewidth = 1, na.rm =
   ↪  TRUE) +
  geom_line(aes(x = year, y = price.house, color = 'house buying'), linewidth = 1,
   ↪  linetype = 2, na.rm = TRUE) +
  geom_line(aes(x = year, y = price.rent, color = 'rent'), linewidth = 1, linetype =
   ↪  3, na.rm = TRUE) + facet_wrap(.~country, nrow = 4, scales = 'free_y') +
   ↪  labs(title = 'Evolución porcentual del salario de los jóvenes, precio de la
   ↪  vivienda y precio del alquiler', subtitle = 'En función de países') +
   ↪  xlab('Year') + ylab('Percentage increase (%)')
```



```
ggsave('results/viviendas_salarios.pdf', width = 14, height = 10, dpi = 300)
```

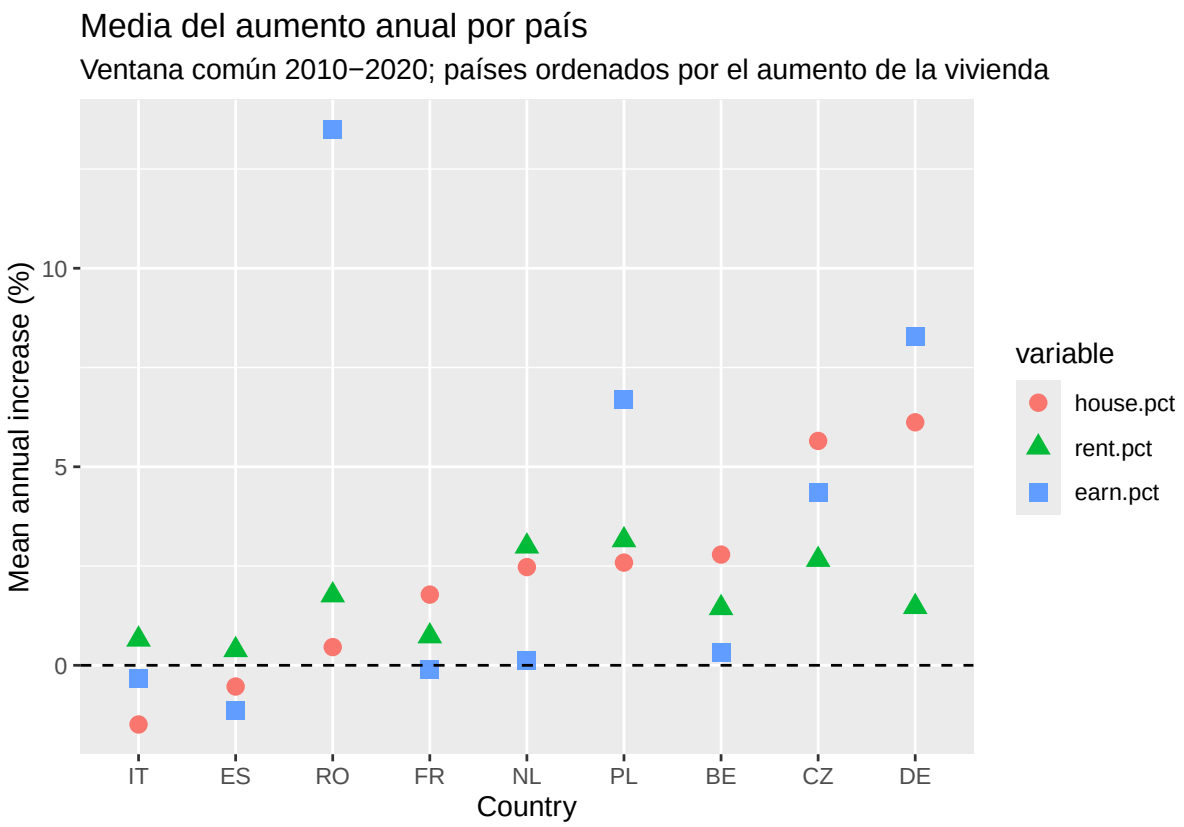
```
ggplot(data = filter(jovenes.df, country != 'EL', year >= 2005 & year <= 2025)) +
  geom_line(aes(x = year, y = earn.prop.house, color = 'by house prices'), linewidth =
   ↪  1, na.rm = TRUE) +
  geom_line(aes(x = year, y = earn.prop.rent, color = 'by rent'), linewidth = 1, na.rm
   ↪  = TRUE) + facet_wrap(.~country, nrow = 4, scales = 'free_y') + labs(title =
   ↪  'Evolución del salario real en proporción al precio de la vivienda y del
   ↪  alquiler', subtitle = 'En función de países') + xlab('Year') + ylab('Percentage
   ↪  increase (%)')
```




```
ggsave('results/proporcion_salario.pdf', width = 14, height = 10, dpi = 300)
```

```
medias.long <- medias.pais %>% pivot_longer(all_of(vars.pct), names_to = 'variable',
 ↪  values_to = 'media') %>% mutate(country = factor(country, levels =
 ↪  as.character(medias.pais$country[order(medias.pais$house.pct)])), variable =
 ↪  factor(variable, levels = vars.pct))

ggplot(medias.long, aes(x = country, y = media, color = variable, shape = variable)) +
 ↪  geom_point(size = 3) + geom_hline(yintercept = 0, linetype = 2) + labs(title =
 ↪  'Media del aumento anual por país', subtitle = paste0('Ventana común ', year.ini,
 ↪  '-', year.fin, '; países ordenados por el aumento de la vivienda')) +
 ↪  xlab('Country') + ylab('Mean annual increase (%)')
```




```
ggsave('results/univ_aumentos_medias.pdf', width = 14, height = 10, dpi = 300)
```